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G.A. EASME/EMFF/2017/1.2.1.12/S1/02/SI2.789635

Co-funded by the EMFF programme
of the European Union

"Sustainable Blue Economy"

DiveSafe

"Integrated system for scientific and environmental underwater surveys, with advanced health & safety features"

DIVE SAFE

WP5: Commercialization, Sustainability & Market-readiness (TRL8)

Product Specification – DS_USBL

Responsible Organization: UnivPM

Version 0.1, Date: 21/12/2020

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Preface

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History of Changes:

Date	Version	History of change
21/12/2020	0.1	First version issued by UnivPM

Associated Documents

- Bill of Materials (Appendix A)
- Software Requirements Specification for Docking Station Software

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SECTION P – PHYSICAL REQUIREMENTS

P1. SCOPE AND OBJECTIVES

This document presents the scooter's internal systems and their integration to make it work with the developed DiveSafe components (underwater tablet, environmental and health sensors and the acoustic localization system). In this case the high-end solution, represented by the Docking Station (DS) connected to the USBL with external antenna, will be described. Hereafter in this specification, this product will be referred to as the DS_USBL.

P2. DESCRIPTION

The new designed internal system integrates and manages all information and data from and to the other elements of the DiveSafe system. In detail, it will be able to manage:

i) the input and output control messages to the sensors;
ii) the data exchange between the underwater tablet and the scooter subsystems/payloads;
iii) the processing of the raw data from the sensors.

The DS_USBL is composed of the main components illustrated in the figure below (Figure 1):

1. Evologics S2CR 18/34 USBL trasponder electronics, inserted in a Derlin link cap
2. Transponder Cap
3. External Antenna with a collar and a 3D-printed support (in the CAD 2 collars are presented)
4. Slide + Electronics
5. DS Hull.

Therefore, the DS_USBL is mainly composed of two cylindrical structures (DS and USBL) and the external antenna.

Regarding its buoyancy, it is slightly positive with the weight of the antenna in front. It must be known that these elements can be installed on both DPVs, even if in this document their installation on the High-End Scooter will be considered.

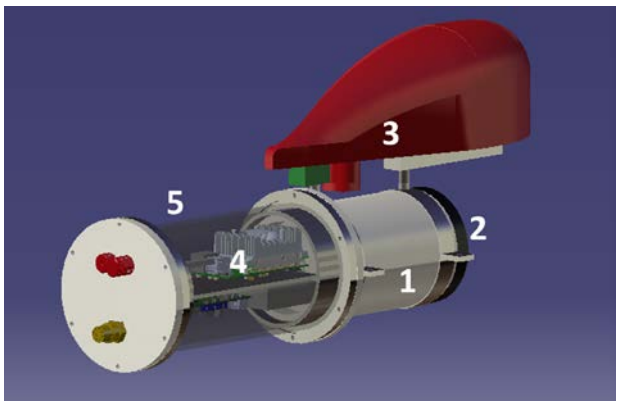

Figure 1: Main components of the DS_USBL

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The solution with USBL antenna, that allows both positioning and communications functionalities, could appear a little bit "bulky". However, having the USBL antenna instead of modem antenna, improves the position precision (it was measured 3m with modem instead of the 1.5m with USBL).

P3. GENERAL REQUIREMENTS

P3.1 Docking Station

The single DS (Figure 2) is composed only of the previous blocks 4 and 5 (Slide + Electronics and DS Hull). The DS scaffold is mainly composed of:

1) Cap
2) Flange
3) Router for WiFi and acoustic communications
4) AHRS
5) OpenUPS2
6) Nvidia Jetson TX2
7) Tubular Plexiglas with outer diameter 120 mm and thickness 5 mm
8) Flange
9) Aluminum plate.

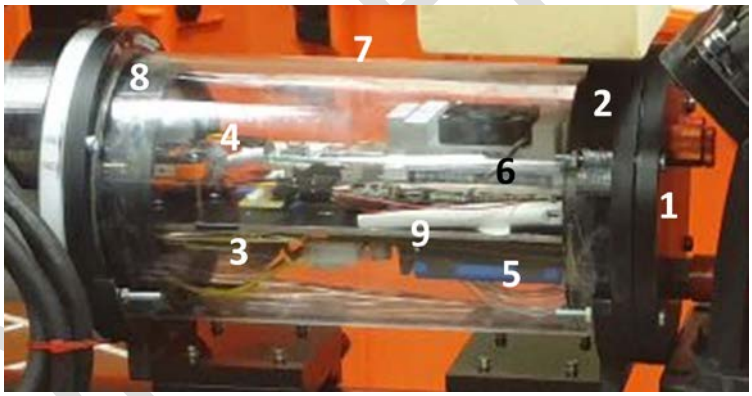

Figure 2. Docking Station

The two aluminum flanges are glued to the plexiglass tube, while the cap, inserted in one flange, is made of Delrin. The cap has two o-rings that guarantee the waterproof. This solution allows to easily insert a slide (in this case composed of an aluminium plate) on which all the necessary electronic components are fixed. Furthermore, its cylindrical shape can be attached to any vehicle while maintaining approximately the same hydrodynamic performance.

In Figure 3, all the electronic components of the DS are detailed and indicated as follows:

1. Nvidia Jetson TX2 kit (also indicated in the previous figure)
2. Auvidea J121
3. OpenUps2 with LiFePO4 Batteries (also indicated in the previous figure)
4. AHRS: Xsens MTi-670 DK (also indicated in the previous figure)

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5. Netgear EX6120 (Router also indicated in the previous figure)
6. MP1584 Buck / Step Down
7. Blue Robotics Depth Meter 30 Bar
8. Enclosure Vent and Plug
9. Subconn 8 pin connector male MCBH8M with its cover cap.

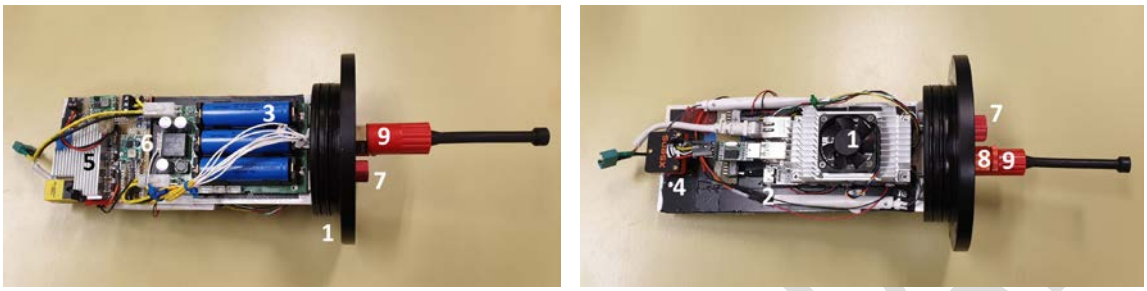

Figure 3. Main components of the Docking Station

P3.2 Complete DS_USBL

The DS_USBL solution (Figure 4) considers the connection of the DS to the USBL with external antenna. In this case, all the electronics of the S2CR 18/34 USBL transponder is contained in an additional Delrin cylinder that represent a link cap, specially designed and produced: the latter is fixed on one side to the housing flange and on the other is closed with transponder cap.

The external Antenna of the USBL is installed on a collar and a 3D-printed support on the Delrin part.

The DS_USBL can be installed on a high-end scooter (in this case, SUEX XJT) by means of two 3D printed collars screwed on the SUEX elastic band video supports: the designed solution could be easily installed on these types of scooter. Non-slip rubbers can be added to avoid even more the DS sliding.

Besides the DS_USBL, the cradle and the tablet housing will be then placed.

Figure 4: DS_USBL above the SUEX scooter

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P4. ENVIRONMENTAL AND MECHANICAL

P4.1 Depth

Max depth for normal operation: 60m

P4.2 Temperature

Max and min temp for normal operation: from -5°C to 35 °C

Max and min temp for storage: from -5°C to 35 °C

P4.3 Components

- Plexiglass Tube
- Aluminum plate
- Steel screws
- Derlin Cylinder
- Aluminium bars.

P5. POWER SUPPLY

Battery type:	18650 LiFePO4 3.2V Batteries
Operating time:	1 hour & 15 min
Charge time:	3 hours

P6. ASSEMBLY

First you need to insert the batteries into the OpenUPS2. Then you have to insert the 2 o-rings previously greased in the appropriate slots of the cap and insert the electronics into the case, making sure to match the orange lines marked. Four screws and nuts are used, two of them are fixed to one of the 3D printed collars screwed on the SUEX elastic band.

At this point, the connector of the USBL transponder must be connected to the DS and the Derlin link cap (already closed on the other side with the transponder cap) must be fixed to the DS through 4 threaded rods, two of them are fixed to one of the 3D printed collars screwed on the SUEX elastic band, whereas the other two threaded rods need to be also fixed to the 3D-printed support of the external antenna, that must be fixed to the Derlin link cap also by means of the collar.

The next step is then to screw the plug into the enclosure vent and finally, the SUEX elastic band video supports can be installed on the scooter.

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P7. CONNECTION

The Docking station has the following connections:

1) Telnet connection via Ethernet connector between Jetson mainboard and modem
2) I2C connection between mainboard and sensors
3) Wi-Fi router to which Mainboard, Diver_MB and Tablet (NA) are connected (TCP and UDP protocols are used)

The DS communication with the surface happens through the acoustic modem.

P7.1 DS Power On

MCBH8M is a waterproof connector (Figure 7) used for the DS power on: in fact, its first ignition must be carried out with the correspondent female connector, specially connected to the power supply.

Figure 5. MCBH8M connector

The pinout of this connector is the following (in the previous figure, you can see a white mark for pins 7 and 8):

1) Jetson power on+
2) Jetson power on-
3) NC
4) NC
5) VIN 13-24 V
6) GND
7) Openups Power on+
8) OpenUps Power on-

The first time the batteries are inserted, or every time they are replaced or have completely discharged, the ignition must be carried out using the female connector. Pins 5 and 6 of this connector

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must be connected to a generator or power supply with voltage in the 13-24 V range, respectively, pin 5 to 13-24V and 6 to ground.

Then you need to turn on the Jetson board by shorting pins 1 and 2 of the power on the connector.

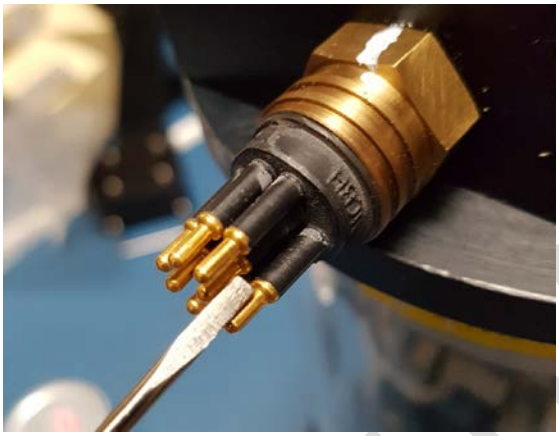

Figure 6. How to turn the Jetson board on

If, on the other hand, the first power-up was made, to switch on the OpenUPS2, simply short pins 7 and 8 as shown in the Figure 7, while the Jetson must be already turned on as explained earlier.

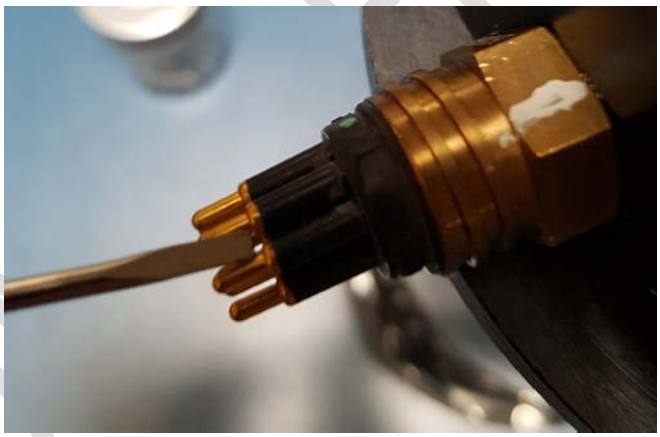

Figure 7. How to switch the OpenUPS2 on

When the Jetson is switched on, the upper fan will start to spin for a few seconds and the LEDs of the Ethernet connector and the rear LED (near the cap) will be on. The fan will restart when the temperature surpasses a certain threshold.

P7.2 Battery Recharge

In order to charge the batteries, the DS must be opened and the batteries must be inserted in their special charger.

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Section S – SURFACE

S1. ACTIVATION

The software inside the Jetson mainboard starts by itself at boot, once the power-on procedure is performed (see P7.1).

Section G – GENERAL

G1. MANUFACTURING DEFAULT SETTINGS

- Water Type: fresh and salt water

G2. GENERAL RECOMMENDATIONS

Every time you have to open or close the Docking Station you have to unscrew the plug of the enclosure vent and remove it to let the air pass: this will make operations easier. Always remember to put the plug back in, especially before entering the water!

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Appendix A – Bill of Materials

- Evologics S2CR 18/34 USBL with the external Antenna
- Derlin link cap, custom made
- Four screws and nuts for DS cap
- Collar for the antenna
- 3D-printed support for the antenna, custom made
- 4 threaded rods with nuts
- Two 3D printed collars for the DS, custom made
- Two SUEX elastic band video supports with 4 screws each one
- DS, composed of:
    - Slide + Electronics
    - DS Hull.

All DS parts will be now listed within the following table, that summarizes the bill of materials for the Docking Station's construction and assembly. For some products the price depends on the seller, so the price is an indication starting from which can be chosen to buy from other distributors. Prices were verified on 22 June 2020. The links show the picture of the component and its distributor.

Product	**Quantity**	**Link**	**Price**	**Tot Price**
Nvidia Jetson TX2 https://www.nvidia.com/it-it/autonomous-machines/embedded-systems/jetson-tx2/kit	1	https://www.nvidia.com/it-it/autonomous-machines/embedded-systems/jetson-tx2/	500	500
Auvidea J121	1	https://auvidea.eu/product/70718/	240	240
Netgear EX6120	2	https://www.netgear.com/home/products/networking/wifi-range-extenders/EX6120.aspx	45	90
Xsens MTi-670 DK	1	https://www.mouser.dk/ProductDetail/Xsens/MTi-670?qs=T3oQrply3y%2FtAuHg3wTCHw%3D%3D	1233.68	1233.68
OpenUps2	1	https://www.cartft.com/catalog/il/1821	109	109

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18650 LiFePO4 Batteries	6	https://it.rs-online.com/web/p/batterie-a-dimensioni-speciali-ricaricabili/8183005/	10	60
MP1584 Buck / Step Down	4	https://robu.in/product/mini-mp1584-dc-dc-adjustable-buck-module-3a/	1	4
Blue Robotics Depth Meter 30 Bar	3	https://bluerobotics.com/store/sensors-sonars-cameras/sensors/bar30-sensor-r1/	64	192
Enclosure Vent and Plug	1	https://bluerobotics.com/store/cables-connectors/penetrators/vent-asm-r1/	8	8
Subconn 8 pin connector male MCBH8M with its cover cap	2	https://www.amronintl.com/seacon-micro-wet-con-male-bulkhead-connector-with-teflon-wire-pigtail-mcbh8m.html	116.44	232.88
Subconn 8 pin connector female MCBH8F	2	https://www.amronintl.com/seacon-micro-wet-con-female-bulkhead-connector-with-teflon-wire-pigtail-mcbh8f.html	116.44	232.88
1 meter Plexyglass Tube 120(e)/ 110(i) mm	1	https://www.plexishop.it/it/plexiglass/tubi/trasparenti/tubo-plexiglass-o-120-e-110-i-mm.html	62	62
O-ring for the Tube (2.62 mm cross-section and 95 mm internal diameter)	4			
Stainless Steel Screw/Bolt Kit	1	https://it.rs-online.com/web/p/kit-viti-e-bulloni/0281984/	65	65
Corner fixing brackets	8	http://www.profilialluminio-online.it/squadra_angolare_di_fissaggio_per_profili_30x30_vendita_121.php	2	16
DIN Type Matrixboard	1	https://it.rs-online.com/web/p/schede-a-matrice/5153301/	13	13

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150x700 mm cylindrical block Polyoxymethylene (acetal resin)	1			
Bulgin, 400 5A Male Solder Circular Connector Contact	32	https://it.rs-online.com/web/p/contatti-per-connettori-circolari/4261322/	0.88	28.16
Bulgin, 400 5A Female Solder Circular Connector Contact	32	https://it.rs-online.com/web/p/contatti-per-connettori-circolari/4261344/	1.67	53.44
2 Way PCB Terminal Strip 2.54 mm	20	https://it.rs-online.com/web/p/blocchetti-terminali-da-pcb/7901098/	0.60	12
2 Way PCB Terminal Block 5 mm	5	https://it.rs-online.com/web/p/blocchetti-terminali-da-pcb/8970803/	0.29	1.45
JST, PHR Female Connector	2	https://it.rs-online.com/web/p/custodie-per-connettori-da-pcb/8201466/	0.07	0.14

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